```
-------------------------------------------------------------------------------
-------------------------------------------------------------
      name:  <unnamed>
       log:  C:\Evaluation course\RCT\ Endline_Analysis.Results.log
  log type:  text
 opened on:  10 Oct 2026, 01:22:19

. ********************************************************************************
. *IMPACT OF ACCESS TO PIPED WATER ON HEALTHCARE EXPENDITURE
. *CAUSAL STUDY OF PIPED WATER INTERVENTION USING RANDOMIZED ASSIGNMENT METHOD
. *ANALYST: MESHACK OMONDI
. ********************************************************************************
.
. ********************************************************************************
. *DATA IMPORT & INITIAL CHECKS
. ********************************************************************************
. import excel using "Randomized_Assignment_Final.xlsx",
sheet("F_Endline_Data(full)") firstrow clear

. count
  1,032

. describe

Contains data
  obs:         1,032
 vars:            17
 size:        71,208
-------------------------------------------------------------------------------
-------------------------------------------------------------
              storage   display    value
variable name   type    format     label      variable label
-------------------------------------------------------------------------------
-------------------------------------------------------------
Household_ID    str6    %9s                   Household_ID
Village         str12   %12s                  Village
Household_Size  byte    %10.0g                Household_Size
Monthly_House~e long    %10.0g                Monthly_Household_Income
Current_Water~e str11   %11s                  Current_Water_Source
Time_Spent_Co~s int     %10.0g                Time_Spent_Collecting_water_Mins
Distance_to_H~y double  %10.0g                Distance_to_Health_facility
Monthly_Healt~e int     %10.0g                Monthly_Health_Expenditure
Treatment       byte    %10.0g                Treatment
Eligible_Endl~e byte    %10.0g                Eligible_Endline
Selected_foll~p byte    %10.0g                Selected_followup
Attrition       byte    %10.0g                Attrition
Random_Number   double  %10.0g                Random_Number
Contacted       byte    %10.0g                Contacted
Non_response    byte    %10.0g                Non_response
Random_NR       double  %10.0g                Random_NR
```

```
Endline_compl~e byte    %10.0g                Endline_complete
-------------------------------------------------------------------------------
-------------------------------------------------------------
Sorted by:
     Note: Dataset has changed since last saved.

. summarize

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
Household_ID |          0
     Village |          0
Household_~e |      1,032    5.574612    2.253622          1         11
Monthly_Ho~e |      1,032    46257.85    26735.62       7600     174600
Current_Wa~e |          0
-------------+---------------------------------------------------------
Time_Spent~s |      1,032    30.63081    30.98253          0        120
Distance_t~y |      1,032    6.759109    3.800719         .2         15
Monthly_He~e |      1,032    5444.767    3965.434          0      19100
   Treatment |      1,032          .5    .5002424          0          1
Eligible_E~e |      1,032           1           0          1          1
-------------+---------------------------------------------------------
Selected_f~p |      1,032           1           0          1          1
   Attrition |      1,032     .120155    .3253004          0          1
Random_Num~r |      1,032     .487613    .2878065   .0000347    .999714
   Contacted |      1,032           0           0          0          0
Non_response |      1,032    .0436047    .2043128          0          1
-------------+---------------------------------------------------------
   Random_NR |        908    .5102301    .2937389   .0006001   .9973363
Endline_co~e |      1,032    .8362403    .3702368          0          1

. codebook

-------------------------------------------------------------------------------
-------------------------------------------------------------
Household_ID
                                                      Household_ID
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  string (str6)

         unique values:  1,032                    missing "":  0/1,032

              examples:  "HH173"
                         "HH379"
                         "HH586"
                         "HH792"

-------------------------------------------------------------------------------
```

```
-------------------------------------------------------------------
Village
                                                          Village
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                 type:  string (str12)

         unique values:  4                        missing "":  0/1,032

            tabulation:  Freq.  Value
                           243  "Kiserian"
                           227  "Namanga"
                           292  "Oloitoktok A"
                           270  "Oloitoktok B"

               warning:  variable has embedded blanks

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Household_Size
                                                   Household_Size
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                 type:  numeric (byte)

                range:  [1,11]                       units:  1
         unique values:  11                       missing .:  0/1,032

                 mean:   5.57461
            std. dev:   2.25362

           percentiles:        10%       25%       50%       75%       90%
                                 3         4         5         7         9

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Monthly_Household_Income
                                        Monthly_Household_Income
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                 type:  numeric (long)

                range:  [7600,174600]                units:  100
         unique values:  569                      missing .:  0/1,032

                 mean:   46257.8
            std. dev:   26735.6
```

```
           percentiles:        10%       25%       50%       75%       90%
                              20000     27600     40250     57550     79400

-------------------------------------------------------------------------------
-------------------------------------------------------------
Current_Water_Source
                                              Current_Water_Source
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  string (str11)

          unique values:  5                       missing "":  0/1,032

             tabulation:  Freq.  Value
                            113  "Borehole"
                            495  "Piped Water"
                            152  "Rainwater"
                            148  "River"
                            124  "Vendor"

                warning:  variable has embedded blanks

-------------------------------------------------------------------------------
-------------------------------------------------------------
Time_Spent_Collecting_water_Mins
                                 Time_Spent_Collecting_water_Mins
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  numeric (int)

                 range:  [0,120]                      units:  1
          unique values:  115                    missing .:  0/1,032

                  mean:   30.6308
             std. dev:   30.9825

           percentiles:        10%       25%       50%       75%       90%
                                 2         5        17        53        72

-------------------------------------------------------------------------------
-------------------------------------------------------------
Distance_to_Health_facility
                                       Distance_to_Health_facility
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  numeric (double)
```

```
                range:  [.2,15]                     units:  .1
        unique values:  149                      missing .:  0/1,032

                 mean:   6.75911
            std. dev:   3.80072

          percentiles:        10%       25%       50%       75%       90%
                              1.7       3.5       6.6       9.7        12

-------------------------------------------------------------------------------
-------------------------------------------------------------
Monthly_Health_Expenditure
                                       Monthly_Health_Expenditure
-------------------------------------------------------------------------------
-------------------------------------------------------------

                 type:  numeric (int)

                range:  [0,19100]                    units:  100
        unique values:  156                      missing .:  0/1,032

                 mean:   5444.77
            std. dev:   3965.43

          percentiles:        10%       25%       50%       75%       90%
                                0      2200      5150      8400     10900

-------------------------------------------------------------------------------
-------------------------------------------------------------
Treatment
                                                       Treatment
-------------------------------------------------------------------------------
-------------------------------------------------------------

                 type:  numeric (byte)

                range:  [0,1]                        units:  1
        unique values:  2                        missing .:  0/1,032

           tabulation:  Freq.  Value
                          516  0
                          516  1

-------------------------------------------------------------------------------
-------------------------------------------------------------
Eligible_Endline
                                                 Eligible_Endline
-------------------------------------------------------------------------------
-------------------------------------------------------------
```

```
                  type:  numeric (byte)

                 range:  [1,1]                        units:  1
         unique values:  1                        missing .:  0/1,032

            tabulation:  Freq.  Value
                         1,032  1

-------------------------------------------------------------------------------
-------------------------------------------------------------
Selected_followup
                                                Selected_followup
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  numeric (byte)

                 range:  [1,1]                        units:  1
         unique values:  1                        missing .:  0/1,032

            tabulation:  Freq.  Value
                         1,032  1

-------------------------------------------------------------------------------
-------------------------------------------------------------
Attrition
                                                    Attrition
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  numeric (byte)

                 range:  [0,1]                        units:  1
         unique values:  2                        missing .:  0/1,032

            tabulation:  Freq.  Value
                           908  0
                           124  1

-------------------------------------------------------------------------------
-------------------------------------------------------------
Random_Number
                                                  Random_Number
-------------------------------------------------------------------------------
-------------------------------------------------------------

                  type:  numeric (double)

                 range:  [.00003471,.99971398]        units:  1.000e-11
```

```
         unique values:  1,032                    missing .:  0/1,032

                  mean:   .487613
             std. dev:   .287806

           percentiles:        10%       25%       50%       75%       90%
                            .09948    .24354   .463324   .729997   .902037

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Contacted
                                                        Contacted
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                  type:  numeric (byte)

                 range:  [0,0]                        units:  1
         unique values:  1                        missing .:  0/1,032

            tabulation:  Freq.  Value
                         1,032  0

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Non_response
                                                     Non_response
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                  type:  numeric (byte)

                 range:  [0,1]                        units:  1
         unique values:  2                        missing .:  0/1,032

            tabulation:  Freq.  Value
                           987  0
                            45  1

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Random_NR
                                                       Random_NR
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                  type:  numeric (double)

                 range:  [.00060015,.99733627]        units:  1.000e-11
         unique values:  908                      missing .:  124/1,032
```

```
                mean:    .51023
             std. dev:   .293739

          percentiles:        10%       25%       50%       75%       90%
                           .101286   .255575   .504258   .770052   .913312

-------------------------------------------------------------------------------
-------------------------------------------------------------------
Endline_complete
                                                  Endline_complete
-------------------------------------------------------------------------------
-------------------------------------------------------------------

                 type:  numeric (byte)

                range:  [0,1]                        units:  1
        unique values:  2                          missing .:  0/1,032

           tabulation:  Freq.  Value
                          169  0
                          863  1

. ************************************************************************
. *Follow-up
. tabulate Treatment

  Treatment |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        516       50.00       50.00
          1 |        516       50.00      100.00
------------+-----------------------------------
      Total |      1,032      100.00

. tabulate Attrition Treatment, row

+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+

           |       Treatment
 Attrition |         0          1 |     Total
-----------+----------------------+----------
         0 |       453        455 |       908
           |     49.89      50.11 |    100.00
-----------+----------------------+----------
         1 |        63         61 |       124
```

```
           |     50.81      49.19 |    100.00
-----------+----------------------+----------
     Total |       516        516 |     1,032
           |     50.00      50.00 |    100.00


. tabulate Non_response Treatment, row

+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+

Non_respon |       Treatment
        se |         0          1 |     Total
-----------+----------------------+----------
         0 |       495        492 |       987
           |     50.15      49.85 |    100.00
-----------+----------------------+----------
         1 |        21         24 |        45
           |     46.67      53.33 |    100.00
-----------+----------------------+----------
     Total |       516        516 |     1,032
           |     50.00      50.00 |    100.00


. ********************************************************************************
. *EDNLINE DESCRIPTIVE STATISTICS
. *Descriptive statistics for Successfully interviewed(863)
.
. summarize Household_Size Monthly_Household_Income ///
> Time_Spent_Collecting_water_Mins ///
> Distance_to_Health_facility ///
> Monthly_Health_Expenditure if Endline_complete == 1

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
Household_~e |        863     5.60139    2.236815          1         11
Monthly_Ho~e |        863    46063.38    26598.66       7600     174600
Time_Spent~s |        863    30.25724     30.6457          0        120
Distance_t~y |        863    6.678216    3.820128         .2         15
Monthly_He~e |        863    5466.049    3970.502          0      19100

.
. * Descriptive statistics by Treatment
. tabstat Household_Size Monthly_Household_Income ///
> Time_Spent_Collecting_water_Mins ///
> Distance_to_Health_facility ///
```

```
> Monthly_Health_Expenditure if Endline_complete == 1, ///
> by(Treatment) statistics(n mean sd)

Summary statistics: N, mean, sd
  by categories of: Treatment (Treatment)

Treatment |  Househ~e  Month~me  Time_S~s  Distan~y  Month~re
----------+---------------------------------------------------
        0 |       432       432       432       432       432
          |  5.414352  46048.38  54.12269  6.615278  6001.389
          |   2.25561  23716.87  25.61687  3.915167   3993.73
----------+---------------------------------------------------
        1 |       431       431       431       431       431
          |  5.788863  46078.42  6.336427  6.741299  4929.466
          |  2.204534   29231.3  8.898944   3.72592  3878.095
----------+---------------------------------------------------
    Total |       863       863       863       863       863
          |   5.60139  46063.38  30.25724  6.678216  5466.049
          |  2.236815  26598.66   30.6457  3.820128  3970.502
-------------------------------------------------------------


.
. *Current water source by treatment
. tab Current_Water_Source Treatment if Endline_complete == 1, row chi2

+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+

Current_Wat |       Treatment
  er_Source |         0          1 |     Total
------------+----------------------+----------
   Borehole |        93          4 |        97
            |     95.88       4.12 |    100.00
------------+----------------------+----------
Piped Water |         0        417 |       417
            |      0.00     100.00 |    100.00
------------+----------------------+----------
  Rainwater |       116          7 |       123
            |     94.31       5.69 |    100.00
------------+----------------------+----------
      River |       116          2 |       118
            |     98.31       1.69 |    100.00
------------+----------------------+----------
     Vendor |       107          1 |       108
            |     99.07       0.93 |    100.00
------------+----------------------+----------
```

```
      Total |       432        431 |       863
            |     50.06      49.94 |    100.00

          Pearson chi2(4) = 809.4258   Pr = 0.000

. ******************************************************************************
.  * 4. CONTINUOUS OUTCOME COMPARISONS
.  ******************************************************************************
.
. ******************************************************************************
. * 4. CONTINUOUS OUTCOME COMPARISONS
. ******************************************************************************
.
. * Primary outcome: monthly healthcare expenditure
. ttest Monthly_Health_Expenditure if Endline_complete == 1, by(Treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     432    6001.389    192.1484     3993.73    5623.724    6379.053
       1 |     431    4929.466    186.8012    3878.095    4562.309    5296.623
---------+--------------------------------------------------------------------
combined |     863    5466.049    135.1575    3970.502    5200.772    5731.325
---------+--------------------------------------------------------------------
    diff |            1071.923    267.9937                545.9251     1597.92
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =   3.9998
Ho: diff = 0                                     degrees of freedom =      861

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0001          Pr(T > t) = 0.0000

.
. * Time spent collecting water
. ttest Time_Spent_Collecting_water_Mins if Endline_complete == 1, by(Treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     432    54.12269    1.232492    25.61687    51.70024    56.54513
       1 |     431    6.336427    .4286471    8.898944    5.493923    7.178931
---------+--------------------------------------------------------------------
combined |     863    30.25724    1.043192     30.6457    28.20975    32.30474
---------+--------------------------------------------------------------------
    diff |            47.78626    1.306093                45.22276    50.34976
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  36.5872
Ho: diff = 0                                     degrees of freedom =      861
```

```
    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 1.0000         Pr(|T| > |t|) = 0.0000          Pr(T > t) = 0.0000

.
. * Monthly household income
. ttest Monthly_Household_Income if Endline_complete == 1, by(Treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     432    46048.38    1141.079    23716.87    43805.61    48291.15
       1 |     431    46078.42    1408.022     29231.3    43310.96    48845.88
---------+--------------------------------------------------------------------
combined |     863    46063.38    905.4288    26598.66    44286.28    47840.49
---------+--------------------------------------------------------------------
    diff |           -30.04264     1811.91                -3586.32    3526.235
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -0.0166
Ho: diff = 0                                     degrees of freedom =      861

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.4934         Pr(|T| > |t|) = 0.9868          Pr(T > t) = 0.5066

.
. * Distance to health facility
. ttest Distance_to_Health_facility if Endline_complete == 1, by(Treatment)

Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     432    6.615278    .1883686    3.915167    6.245042    6.985513
       1 |     431    6.741299    .1794712     3.72592    6.388549    7.094049
---------+--------------------------------------------------------------------
combined |     863    6.678216    .1300387    3.820128    6.422986    6.933445
---------+--------------------------------------------------------------------
    diff |           -.1260215     .260193               -.6367084    .3846654
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -0.4843
Ho: diff = 0                                     degrees of freedom =      861

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.3141         Pr(|T| > |t|) = 0.6283          Pr(T > t) = 0.6859

.
. * Household size
. ttest Household_Size if Endline_complete == 1, by(Treatment)
```

```
Two-sample t test with equal variances
------------------------------------------------------------------------------
   Group |     Obs        Mean    Std. Err.   Std. Dev.   [95% Conf. Interval]
---------+--------------------------------------------------------------------
       0 |     432    5.414352    .1085231     2.25561    5.201052    5.627652
       1 |     431    5.788863    .1061887    2.204534     5.58015    5.997576
---------+--------------------------------------------------------------------
combined |     863     5.60139    .0761421    2.236815    5.451945    5.750836
---------+--------------------------------------------------------------------
    diff |           -.3745113    .1518371               -.6725255    -.076497
------------------------------------------------------------------------------
    diff = mean(0) - mean(1)                                      t =  -2.4665
Ho: diff = 0                                     degrees of freedom =      861

    Ha: diff < 0                 Ha: diff != 0                 Ha: diff > 0
 Pr(T < t) = 0.0069         Pr(|T| > |t|) = 0.0138          Pr(T > t) = 0.9931

.
.
. ******************************************************************************
. * 5. DATA QUALITY AND MISSING VALUE CHECKS
. ******************************************************************************
.
. * Check household identifiers
. duplicates report Household_ID

Duplicates in terms of Household_ID

--------------------------------------
   copies | observations       surplus
----------+---------------------------
        1 |         1032             0
--------------------------------------

.
. * Check missing values in analysis variables
. misstable summarize Household_Size ///
>     Monthly_Household_Income ///
>     Current_Water_Source ///
>     Time_Spent_Collecting_water_Mins ///
>     Distance_to_Health_facility ///
>     Monthly_Health_Expenditure ///
>     Treatment Endline_complete
(variables nonmissing or string)

.
. * Count completed endline interviews
. count if Endline_complete == 1
  863
```

```
.
. * Count completed interviews with observed primary outcome and treatment
. count if Endline_complete == 1 ///
>     & !missing(Monthly_Health_Expenditure, Treatment)
  863

.
. * Examine the distribution of healthcare expenditure
. summarize Monthly_Health_Expenditure if Endline_complete == 1, detail

                Monthly_Health_Expenditure
-------------------------------------------------------------
      Percentiles      Smallest
 1%            0              0
 5%            0              0
10%            0              0       Obs                 863
25%         2300              0       Sum of Wgt.         863

50%         5200                      Mean           5466.049
                        Largest       Std. Dev.      3970.502
75%         8400          16300
90%        10800          17300       Variance       1.58e+07
95%        12300          18700       Skewness       .4277244
99%        15200          19100       Kurtosis       2.552224

.
.
. ******************************************************************************
. * 6. UNADJUSTED TREATMENT EFFECTS
. ******************************************************************************
.
. * Primary outcome: healthcare expenditure
. regress Monthly_Health_Expenditure Treatment ///
>     if Endline_complete == 1, vce(robust)

Linear regression                               Number of obs     =        863
                                                F(1, 861)         =      16.00
                                                Prob > F          =     0.0001
                                                R-squared         =     0.0182
                                                Root MSE          =     3936.4

------------------------------------------------------------------------------
             |               Robust
Monthly_He~e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |  -1071.923   267.9846    -4.00   0.000    -1597.902    -545.943
       _cons |   6001.389   192.1487    31.23   0.000     5624.254    6378.524
------------------------------------------------------------------------------

.
```

```
. * Secondary outcome: time spent collecting water
. regress Time_Spent_Collecting_water_Mins Treatment ///
>     if Endline_complete == 1, vce(robust)

Linear regression                               Number of obs     =        863
                                                F(1, 861)         =    1341.06
                                                Prob > F          =     0.0000
                                                R-squared         =     0.6086
                                                Root MSE          =     19.184

------------------------------------------------------------------------------
             |               Robust
Time_Spent~s |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |  -47.78626   1.304906   -36.62   0.000    -50.34743   -45.22509
       _cons |   54.12269   1.232494    43.91   0.000     51.70364    56.54173
------------------------------------------------------------------------------

.
. * Secondary outcome: household income
. regress Monthly_Household_Income Treatment ///
>     if Endline_complete == 1, vce(robust)

Linear regression                               Number of obs     =        863
                                                F(1, 861)         =       0.00
                                                Prob > F          =     0.9868
                                                R-squared         =     0.0000
                                                Root MSE          =      26614

------------------------------------------------------------------------------
             |               Robust
Monthly_Ho~e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |   30.04264   1812.343     0.02   0.987    -3527.084    3587.169
       _cons |   46048.38    1141.08    40.36   0.000     43808.76       48288
------------------------------------------------------------------------------

.
. * Secondary outcome: distance to health facility
. regress Distance_to_Health_facility Treatment ///
>     if Endline_complete == 1, vce(robust)

Linear regression                               Number of obs     =        863
                                                F(1, 861)         =       0.23
                                                Prob > F          =     0.6282
                                                R-squared         =     0.0003
                                                Root MSE          =     3.8218

------------------------------------------------------------------------------
             |               Robust
```

```
Distance_t~y |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |   .1260215   .2601781     0.48   0.628    -.3846361    .6366791
       _cons |   6.615278   .1883688    35.12   0.000     6.245562    6.984994
------------------------------------------------------------------------------

.
. * Secondary outcome: household size
. regress Household_Size Treatment ///
>     if Endline_complete == 1, vce(robust)

Linear regression                               Number of obs     =        863
                                                F(1, 861)         =       6.08
                                                Prob > F          =     0.0138
                                                R-squared         =     0.0070
                                                Root MSE          =     2.2302

------------------------------------------------------------------------------
             |               Robust
Household_~e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |   .3745113   .1518331     2.47   0.014     .0765049    .6725176
       _cons |   5.414352   .1085232    49.89   0.000     5.201351    5.627353
------------------------------------------------------------------------------

.
.
. ******************************************************************************
. * 7. CATEGORICAL WATER-SOURCE ANALYSIS
. ******************************************************************************
.
. * Examine water-source distribution by treatment group
. tabulate Current_Water_Source Treatment ///
>     if Endline_complete == 1, row chi2

+----------------+
| Key            |
|----------------|
|   frequency    |
| row percentage |
+----------------+

Current_Wat |       Treatment
  er_Source |         0          1 |     Total
------------+----------------------+----------
   Borehole |        93          4 |        97
            |     95.88       4.12 |    100.00
------------+----------------------+----------
Piped Water |         0        417 |       417
            |      0.00     100.00 |    100.00
```

```
------------+----------------------+----------
  Rainwater |       116          7 |       123
            |     94.31       5.69 |    100.00
------------+----------------------+----------
      River |       116          2 |       118
            |     98.31       1.69 |    100.00
------------+----------------------+----------
     Vendor |       107          1 |       108
            |     99.07       0.93 |    100.00
------------+----------------------+----------
      Total |       432        431 |       863
            |     50.06      49.94 |    100.00

          Pearson chi2(4) = 809.4258   Pr = 0.000

.
. * Inspect the type and categories of the water-source variable
. describe Current_Water_Source

              storage   display    value
variable name   type    format     label      variable label
------------------------------------------------------------------------------
------------------------------------------------------------
Current_Water~e str11   %11s                  Current_Water_Source

. tabulate Current_Water_Source if Endline_complete == 1, missing

Current_Wat |
  er_Source |      Freq.     Percent        Cum.
------------+-----------------------------------
   Borehole |         97       11.24       11.24
Piped Water |        417       48.32       59.56
  Rainwater |        123       14.25       73.81
      River |        118       13.67       87.49
     Vendor |        108       12.51      100.00
------------+-----------------------------------
      Total |        863      100.00

.
. * If Current_Water_Source is a STRING variable, encode it first:
. * encode Current_Water_Source, gen(Water_Source_Code)
.
. * If it is numeric and has three or more unordered categories,
. * run the appropriate multinomial logistic regression:
. * mlogit Current_Water_Source Treatment ///
> *     if Endline_complete == 1, rrr
.
. * If you encoded a string variable, use this instead:
. * mlogit Water_Source_Code Treatment ///
> *     if Endline_complete == 1, rrr
```

```
.
.
. ********************************************************************************
. * 8. PRIMARY OUTCOME SENSITIVITY ANALYSIS
. ********************************************************************************
.
. * Compare healthcare expenditure across treatment groups
. tabstat Monthly_Health_Expenditure ///
>     if Endline_complete == 1, ///
>     by(Treatment) statistics(n mean sd p50 p25 p75 min max)

Summary for variables: Monthly_Health_Expenditure
     by categories of: Treatment (Treatment)

Treatment |         N      mean        sd       p50       p25       p75       min
   max
----------+--------------------------------------------------------------------------
-------
        0 |       432  6001.389   3993.73      5850      3000      8850         0
 19100
        1 |       431  4929.466  3878.095      4600      1500      7700         0
 17300
----------+--------------------------------------------------------------------------
-------
    Total |       863  5466.049  3970.502      5200      2300      8400         0
 19100
--------------------------------------------------------------------------------------
-------


.
. * Identify the 99th percentile of healthcare expenditure
. _pctile Monthly_Health_Expenditure ///
>     if Endline_complete == 1, p(99)

.
. local p99 = r(r1)

.
. * Re-estimate the treatment effect excluding values above the 99th percentile
. regress Monthly_Health_Expenditure Treatment ///
>     if Endline_complete == 1 ///
>     & Monthly_Health_Expenditure <= `p99', vce(robust)

Linear regression                               Number of obs     =        855
                                                F(1, 853)         =      15.51
                                                Prob > F          =     0.0001
                                                R-squared         =     0.0179
                                                Root MSE          =     3801.1

------------------------------------------------------------------------------
```

```
             |               Robust
Monthly_He~e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
   Treatment |  -1023.769   259.9935    -3.94   0.000    -1534.071   -513.4668
       _cons |   5873.068   185.4228    31.67   0.000      5509.13    6237.006
------------------------------------------------------------------------------

.
. * 9. ENDLINE GRAPH: HEALTHCARE EXPENDITURE BY TREATMENT GROUP
.
. graph bar (mean) Monthly_Health_Expenditure ///
> if Endline_complete == 1, ///
> over(Treatment) ///
> title("Mean Monthly Healthcare Expenditure at Endline") ///
> subtitle("Treatment vs Control Groups") ///
> ytitle("Mean Monthly Healthcare Expenditure") ///
> ylabel(, angle(horizontal) format(%12.0fc)) ///
> blabel(bar, format(%9.1f)) ///
>

.
. graph export "Healthcare_Expenditure_Endline.png", replace width(2000)
(note: file Healthcare_Expenditure_Endline.png not found)
(file Healthcare_Expenditure_Endline.png written in PNG format)

.
.
. ******************************************************************************
. * 10. SAVE RESULTS AND CLOSE LOG
. ******************************************************************************
.
. log close
      name:  <unnamed>
       log:  C:\Evaluation course\RCT\ Endline_Analysis.Results.log
  log type:  text
 closed on:  10 Oct 2026, 01:22:25
-------------------------------------------------------------------------------
-------------------------------------------------------------
```